

3D Printer Modal Analysis

Final Design Project

Dylan Qiu, Joshua Eah, David Brokhin, Jake Peyser

ME 301 – Mechanical Vibrations

Professor Martin Lawless

May 14, 2026

Contents

1	Introduction	2
1.1	Background and Motivation	2
1.2	Theoretical Background	2
2	Problem Statement	3
2.1	Schematic	3
2.2	Mathematical Description	3
2.3	Fourier Analysis	4
3	Experimental Set Up	5
3.1	Instrumentation	5
3.2	Nodes	6
3.3	Accelerometer Position	6
3.3.1	Center	6
3.3.2	Displaced	7
4	Results (Accelerometer at Node 3)	7
4.1	Raw Data	7
4.2	FFT	8
4.3	Natural Frequencies	8
4.4	Mode Shapes	11
5	Results (Accelerometer at Node 3&4 FAILURE)	12
5.1	Raw Data	12
5.2	FFT	13
5.3	Natural Frequencies	14
6	Discussion	15
6.1	Important Observations	15
6.1.1	Natural Frequencies	15
6.1.2	Mode Shapes	16
6.2	Potential Reasons for Poor Secondary Measurements	16
6.3	Applications	16
7	Conclusion	17
8	Appendix	18
8.1	MATLAB Code	18

1 Introduction

1.1 Background and Motivation

Advancements in FDM printing technology and greatly increased the speed at which 3D printers operate. A far cry from the rudimentary mechanized hot guns injecting plastic onto a moving glass bed, modern printers operate at very high speeds commonly in excess of 600mm/s with accelerations over 20Gs¹. The increasing number of components and sensors on toolheads have made them heavier, and with the increased mass and acceleration print defects due to such vibratory effects have become a problem many manufacturers have developed and are developing solutions for.

The standard solution to printhead resonance in common open-source firmware bases like Klipper is input shaper, which is a resonance avoidance algorithm. Input shaper relies on a sine sweep of the print head's movement with an accelerometer to determine resonance frequencies, then transforms printer movement commands during regular operation to avoid problematic frequencies.

To better understand the resonance characteristics of 3D printers, modal analysis of an 3D printer was conducted using an impact hammer and accelerometer.

1.2 Theoretical Background

Upon neglecting gravity, we assume that the beam attached to the 3D printer obeys the damped Euler-Bernoulli Beam Equation of the following form:

$$\rho AL \frac{\partial^2 w}{\partial t^2} + R \frac{\partial w}{\partial t} + EIL \frac{\partial^4 w}{\partial x^4} = 0 \quad (1)$$

Note that ρ is the density, A is the cross-sectional area, L is the length, E is the Young's Modulus, I is the area moment of inertia, and R is the damping coefficient per unit length of the beam. The variable w denotes the transverse displacement of the beam as a function of space and time. Later, we will implement a forcing element.

This formulation describes a continuous beam with an infinite number of degrees of freedom associated with it as the above PDE can be characterized as a continuum of ODEs for each mode.

In this regard, the system can be treated as an infinite degree of freedom system, and can be analyzed accordingly with modal analysis. Moreover, we wish break down the system into sub-problems of the following form:

$$\ddot{y} + 2\zeta\omega\dot{y} + \omega^2 y = 0, \quad (2)$$

where y is some displacement function, ζ is some damping ratio, and ω is some natural frequency associated with the system.

2 Problem Statement

2.1 Schematic

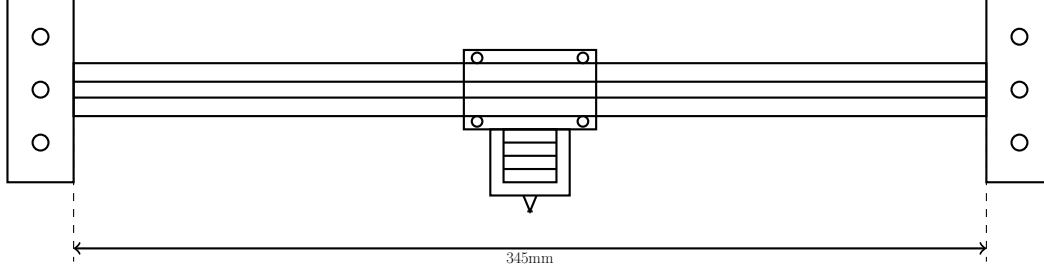


Figure 1: X-beam

Figure 1 illustrates the overall configuration of the X-axis on the printer tested from the side view. This configuration is common across most commercially available FDM printers. The X-axis consists of a standard 2020 profile aluminum extrusion bolted to rigid joints on each end of the beam. The toolhead rides along the extrusion via a carriage assembly.

2.2 Mathematical Description

As eluded to previously, the 3D printer beam can be modeled as the following infinite degree of freedom system:

$$w(x, t) = \sum_{n=1}^{\infty} \phi_n(x) y_n(t) \quad (3)$$

Here we perform separation of variables in space and time. Specifically, $\phi_n(x)$ denotes the n^{th} spatial mode component (dimensionless) and $y_n(t)$ denotes the n^{th} temporal mode component (length units) of the overall transverse displacement $w(x, t)$. Note that the above equation is an orthogonal decomposition. That is, $\{\phi_n\}$ is an orthogonal basis under the L^2 inner product.

Upon plugging this assumption into the governing equation of motion:

$$\rho A L \sum_{n=1}^{\infty} \phi_n(x) \ddot{y}_n(t) + R \sum_{n=1}^{\infty} \phi_n(x) \dot{y}_n(t) + E I L \sum_{n=1}^{\infty} \phi_n''''(x) y_n(t) = 0 \quad (4)$$

which reduces to...

$$\sum_{n=1}^{\infty} \phi_n(x) \left(\rho A \ddot{y}_n(t) + \frac{R}{L} \dot{y}_n(t) + E I \frac{\phi_n''''(x)}{\phi_n(x)} y_n(t) \right) = 0 \quad (5)$$

Because the equation is separable, there necessarily exists $\{\lambda_n\}$ such that $\phi_n''''(x) = \lambda_n \phi_n(x)$, which implies:

$$\sum_{n=1}^{\infty} \phi_n(x) \left(\rho A \ddot{y}_n(t) + \frac{R}{L} \dot{y}_n(t) + E I \lambda_n y_n(t) \right) = 0 \quad (6)$$

Finally, because $\{\phi_n\}$ is an orthogonal basis, we may conclude:

$$\rho A \ddot{y}_n(t) + \frac{R}{L} \dot{y}_n(t) + EI \lambda_n y_n(t) = 0 \quad (7)$$

Moreover, using some slick algebra...

$$\ddot{y}_n + 2\zeta_n \omega_n \dot{y}_n + \omega_n^2 y_n = 0 \quad \text{s.t.} \quad \omega_n = \sqrt{\frac{EI \lambda_n}{\rho A}} \quad \text{and} \quad \zeta_n = \frac{R}{2\rho A L \omega_n} \quad (8)$$

At this point, for each (n^{th}) mode, the transverse displacement can be modeled as a damped oscillator according to the equation above. Each mode has a corresponding natural frequency (ω_n) and damping ratio (ζ_n) which can be determined experimentally.

If not clear, at each position along the beam, it is the superposition of the y_n 's that can be measured with respect to time. From there, Fourier analysis can be used to determine the respective mode shapes and other parameters if needed.

2.3 Fourier Analysis

Because the beam is hit with an impact hammer, an inhomogeneous forcing term $F(x, t) = S_0 \delta(x - x_0) \delta(t - t_0)$ can be added to the RHS of the damped Euler-Bernoulli beam equation, where x_0, t_0 are the respective impact position and time. Furthermore, using similar analysis as in the previous section, $F(x, t)$ can be decomposed over each mode as:

$$F_n(t) = S_0 \phi_n(x_0) \delta(t - t_0) \quad (9)$$

In this regard, we can modify the EOM for each mode as such:

$$\ddot{y}_n + 2\zeta_n \omega_n \dot{y}_n + \omega_n^2 y_n = u_n \delta(t - t_0) \quad \text{for some constant} \quad u_n \quad (10)$$

Upon applying the Fourier transform, where $\mathcal{F}_t \{y_n(t)\}(\omega) = \hat{y}_n(\omega)$:

$$-\left(\frac{\omega}{\omega_n}\right)^{-2} \left(1 - \left(\frac{\omega}{\omega_n}\right)^2 + 2\zeta_n \left(\frac{\omega}{\omega_n}\right) i\right) (-\omega^2 \hat{y}_n(\omega)) = u_n e^{-i\omega t_0} \quad (11)$$

We then define the transfer function (FRF) as the ratio of the Fourier transform of the beam's transverse acceleration to the specific force of the impact hammer:

$$H_n(\omega) = \frac{-\omega^2 \hat{y}_n(\omega)}{u_n e^{-i\omega t_0}} = \frac{-\left(\frac{\omega}{\omega_n}\right)^2}{1 - \left(\frac{\omega}{\omega_n}\right)^2 + 2\zeta_n \left(\frac{\omega}{\omega_n}\right) i} \quad (12)$$

The FRF is thus given by:

$$H(x, \omega) = \sum_{n=1}^{\infty} \phi_n(x) H_n(\omega) = \sum_{n=1}^{\infty} \frac{-\left(\frac{\omega}{\omega_n}\right)^2 \phi_n(x)}{1 - \left(\frac{\omega}{\omega_n}\right)^2 + 2\zeta_n \left(\frac{\omega}{\omega_n}\right) i} \quad (13)$$

Lastly, we take the imaginary part of the FRF for our analysis:

$$H_{\text{imag}}(x, \omega) = \text{Im}(H(x, \omega)) = \sum_{n=1}^{\infty} \frac{2\zeta_n \phi_n(x) \left(\frac{\omega}{\omega_n}\right)^3}{\left(1 - \left(\frac{\omega}{\omega_n}\right)^2\right)^2 + \left(2\zeta_n \left(\frac{\omega}{\omega_n}\right)\right)^2} \quad (14)$$

Depending on the sizes of $\phi_n(x)$, for any fixed position x , the (imaginary part of the) FRF will attain a noticeable peak at the respective natural frequency mode position ω_n . In particular, it will obtain (roughly) a peak value of:

$$H_{\text{imag}}(x, \omega_n) \approx \phi_n(x) \quad (15)$$

It is this fact that allows us to determine both the dominant natural frequencies of the system and the respective spatial-modal amplitudes for the mode shapes.

The mode shape as a function of position (x) can thus be given by:

$$m(x, t) = \sum_{n=1}^{\infty} \phi_n(x) e^{i\omega_n t} \quad (16)$$

Note that for the purposes of the experiment being presented in this report, we only analyzed 5 positions $x = x_1, x_2, x_3, x_4, x_5$, which correspond to the mode shapes nodes $m_i(t) = m(x_i, t)$ for $i = 1, 2, 3, 4, 5$.

3 Experimental Set Up

3.1 Instrumentation

- Impact Hammer
 - PCB Piezotronics 086C03
 - Sensitivity: 2.23 mV/N
- DAQ
 - National Instruments USB-4431 AC/DC converter
 - 4 input, 1 output
- Accelerometer
 - PCB Piezotronics 333B31 SN 37008
 - Sensitivity: 98.5 mV/g, Bias: 11.9V
- Laptop
 - Lenovo ThinkPad
 - MATLAB 2021b, NI Driver Packages
 - The sampling frequency was set to 1000 Hz, because this is well above the frequencies at which 3D printers vibrate at, so it is deemed to be sufficient.

3.2 Nodes

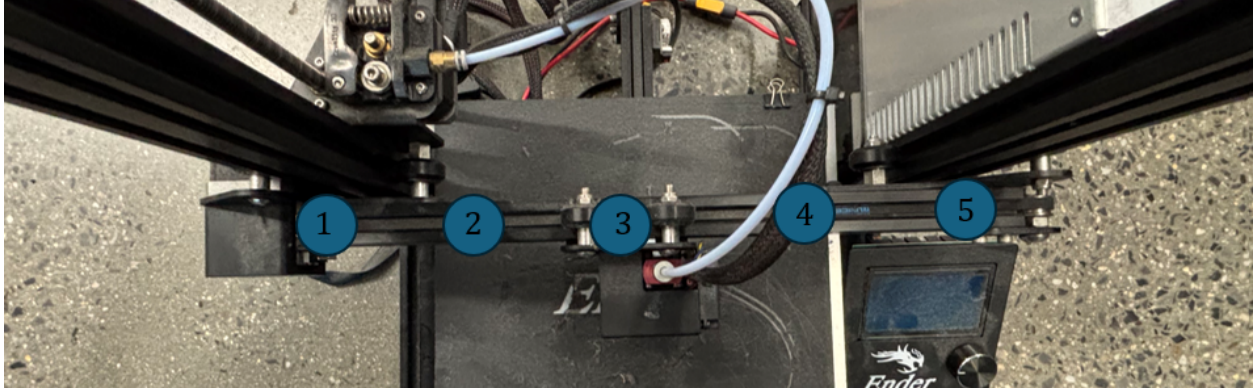


Figure 2: Diagram of impacted and analyzed nodal positions

Figure 2 shows the locations where impact was made by the impact hammer. The points are denoted 1-5 from left to right with respect to the coordinate system of the printer. Each point was evenly spaced through the length of the beam such that position 3 was located at the center of the beam. Impact was made at each location for all testing.

3.3 Accelerometer Position

3.3.1 Center

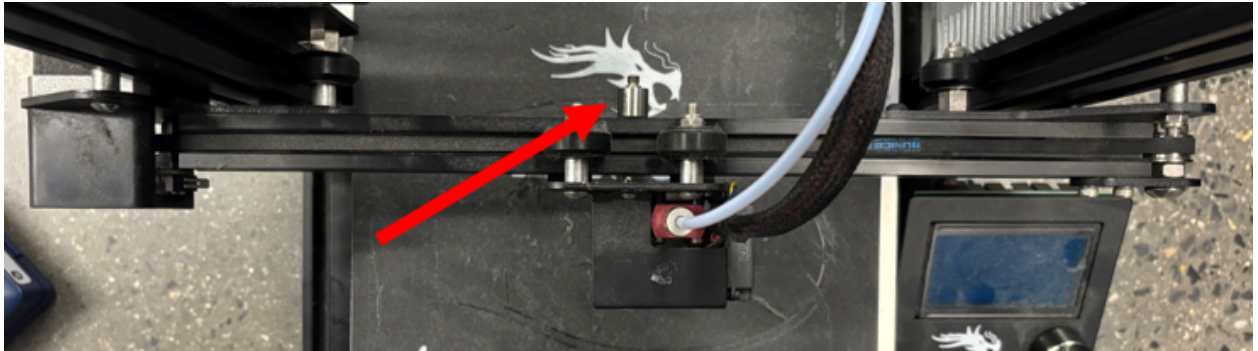


Figure 3: Diagram of Accelerometer Placed at Node 3

Impact testing was conducted with the accelerometer affixed to the center of the beam. Figure 3 shows the position of the accelerometer in this configuration.

3.3.2 Displaced

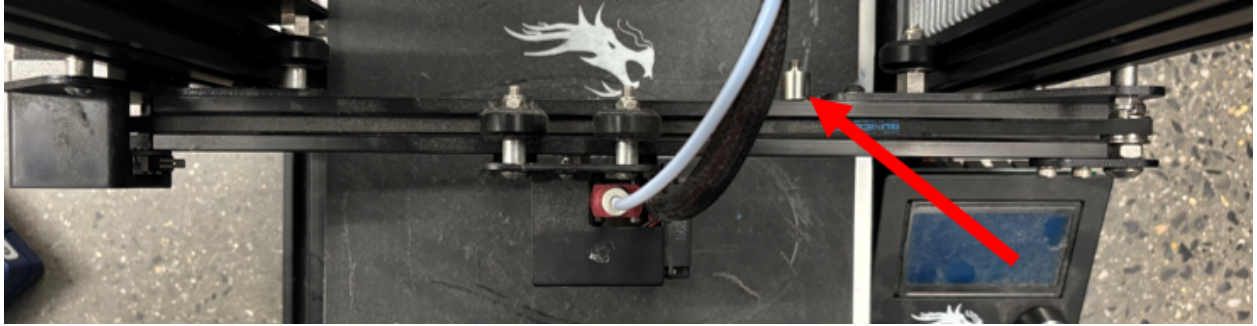


Figure 4: Diagram of Accelerometer Placed at Node 4

Impact testing was also conducted with the accelerometer in a displaced position to better characterize the characteristics of the beam. The displaced position of the accelerometer is shown in Figure 4, located at position 4.

4 Results (Accelerometer at Node 3)

4.1 Raw Data

This is the raw hammer strike data, which is expected to show one peak from the hammer that indicates one clean hit. The following two plots are showing one strike at a specific node (3).

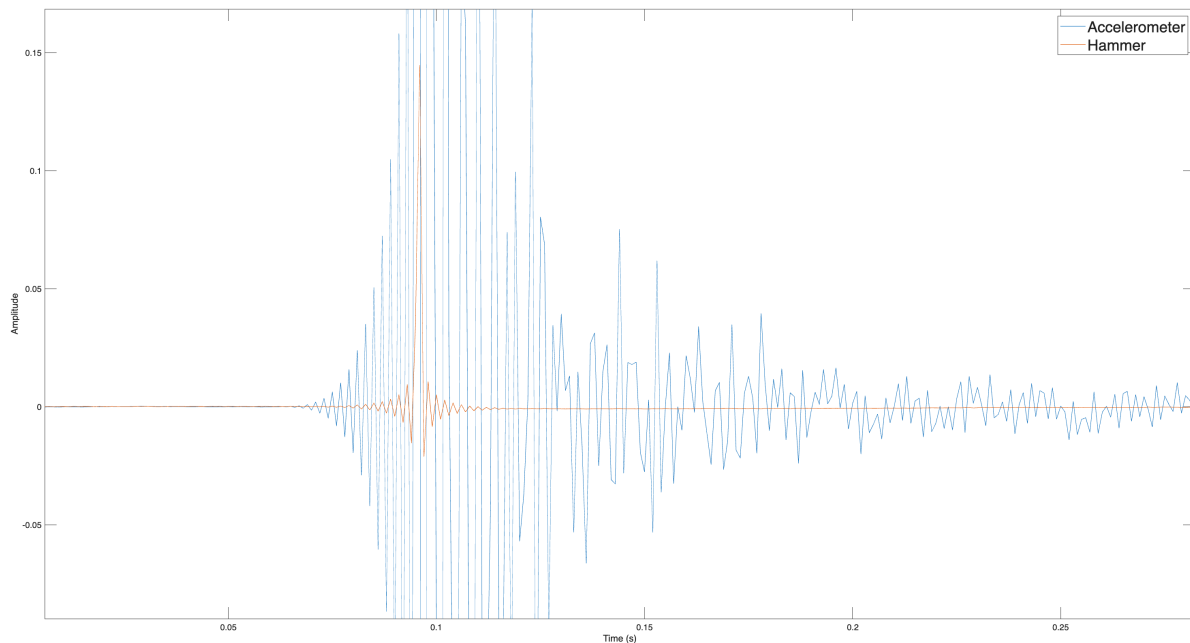


Figure 5: Raw DAQ data. Showing the amplitude of the hammer strike in orange and accelerometer response in blue.

This is force and acceleration as a function of time. The peak is clearly observable right before 0.1 seconds.

4.2 FFT

Taking the raw data, and performing a fast Fourier transform (FFT) gave the following figure.

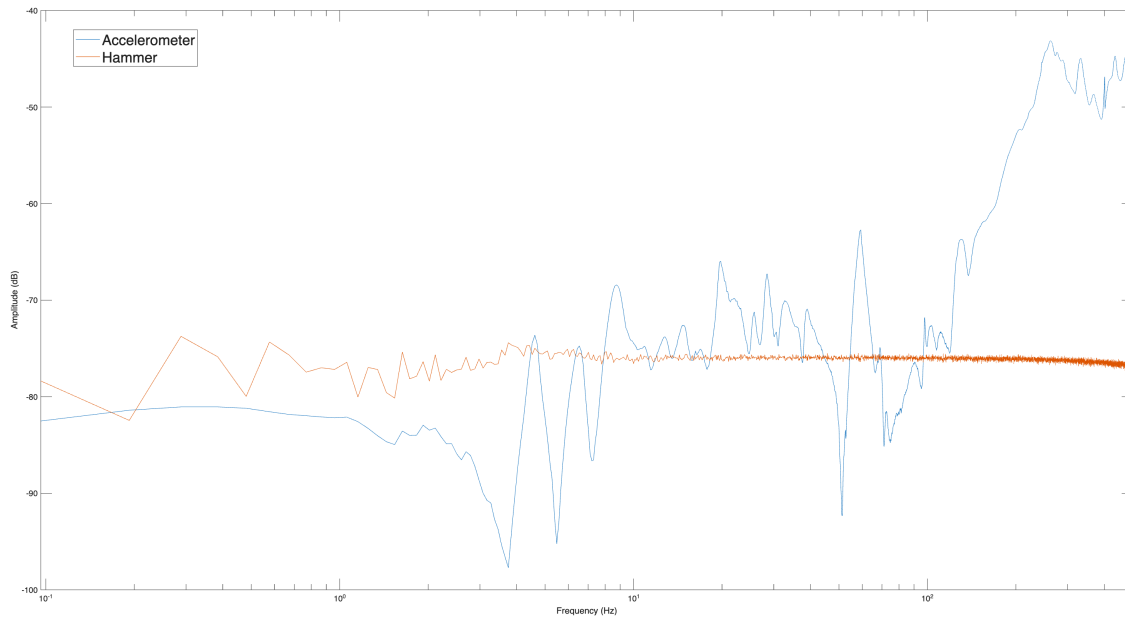


Figure 6: FFT Frequency Graph

This is the force (in orange) and acceleration (in blue) as a function of frequency.

4.3 Natural Frequencies

The positive imaginary part of the experimental FRF for all five nodes is shown below.

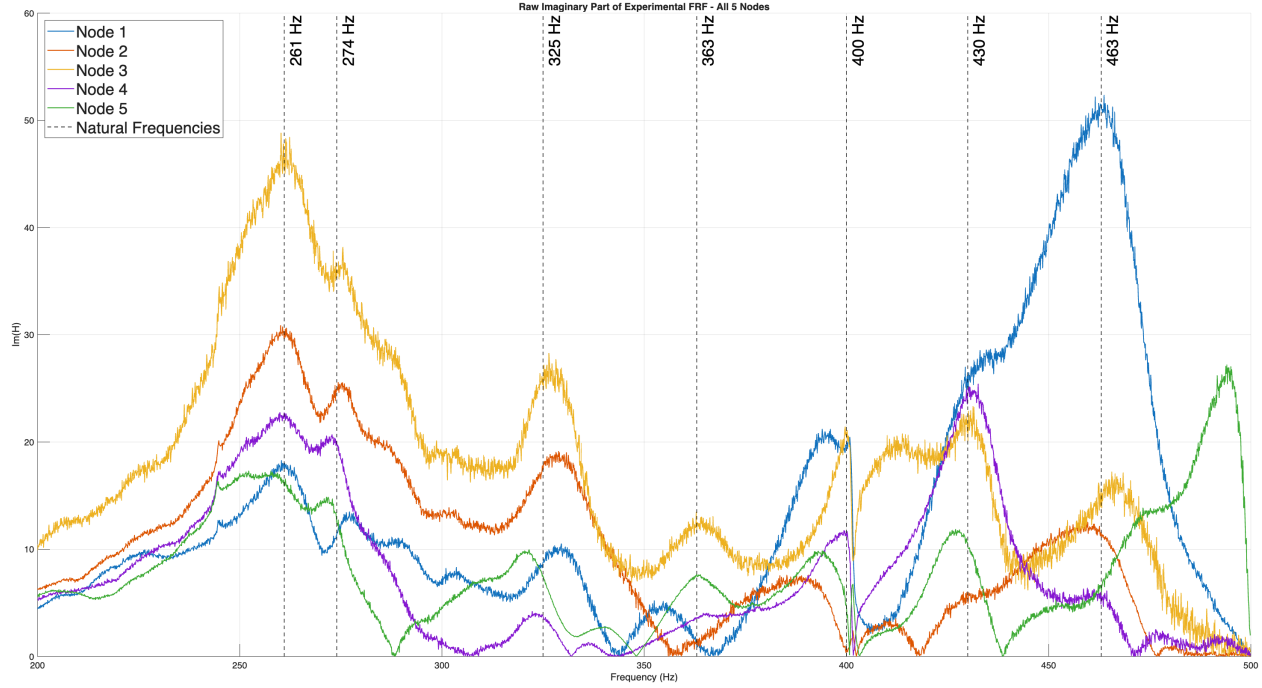


Figure 7: Positive Imaginary Part of Force Response Function

In figure 7 the peaks are demarcated by the black dashed lines are the natural frequencies. Each of the different colors represent the FRF from data collected at each numbered node.

The figure 8 shows the data analysis procedure used for all five nodes. First, identify the peaks from the positive part of FRF in red and write down the frequencies, which are the natural frequencies. Then find the amplitude at these natural frequencies from the whole curve in blue because the amplitudes can be either positive or negative, which are the mode shapes in the following section.

Experimental FRF - All 5 Nodes

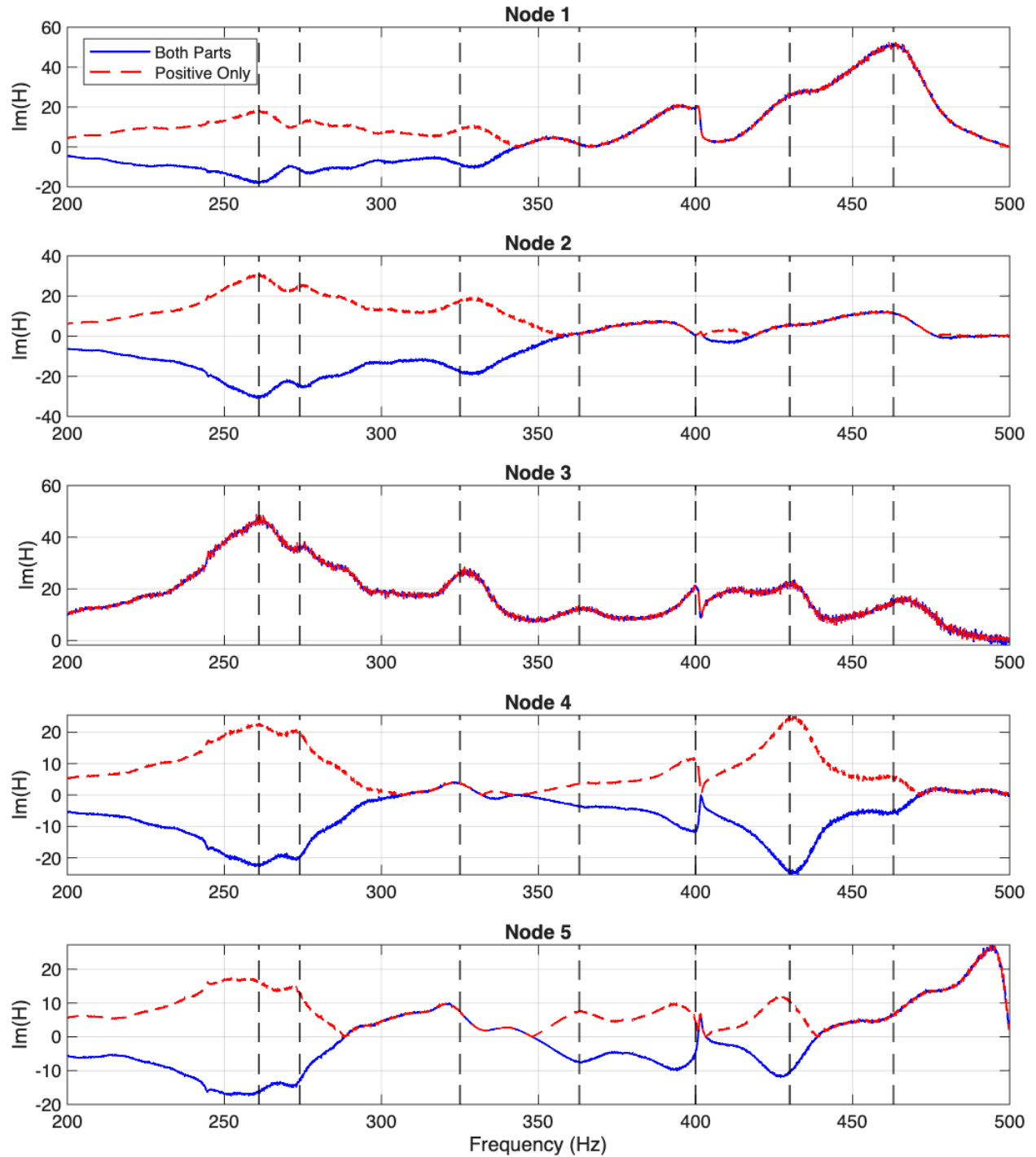


Figure 8: All 5 imaginary FRFs. The red line show the positive part of the curve and the blue line show both positive and negative.

The summary of all the natural frequencies are shown below.

Mode Number	Natural Frequency (Hz)
1	261
2	274
3	325
4	363
5	400
6	430
7	463

4.4 Mode Shapes

The mode shapes for all seven modes are the following. The seven nodes are the 5 tested nodes and the two fixed supports with zero displacement at either end.

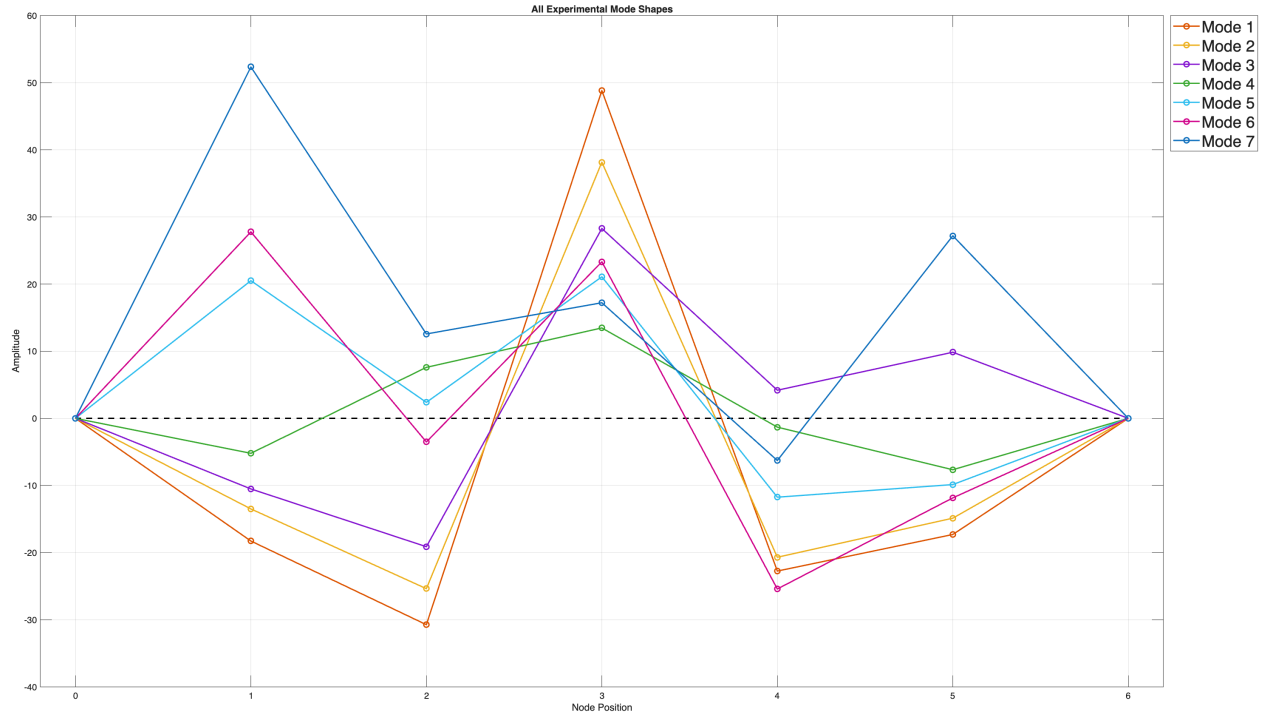


Figure 9: All Experimental Mode Shapes

An example of an individual mode shape is shown in Figure 10:

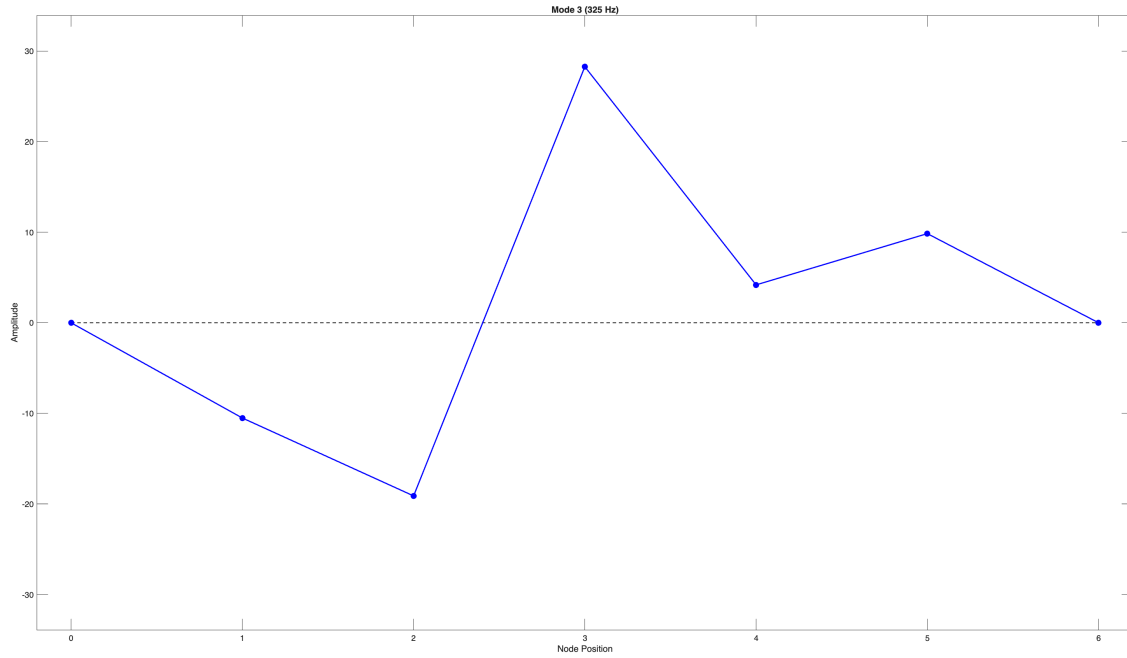


Figure 10: Third Mode Shape

5 Results (Accelerometer at Node 3&4 FAILURE)

5.1 Raw Data

This is the raw hammer strike graph, showing the accelerometer and hammer data. The force is a lot less than the good data set, which could be why the errors showed up later.

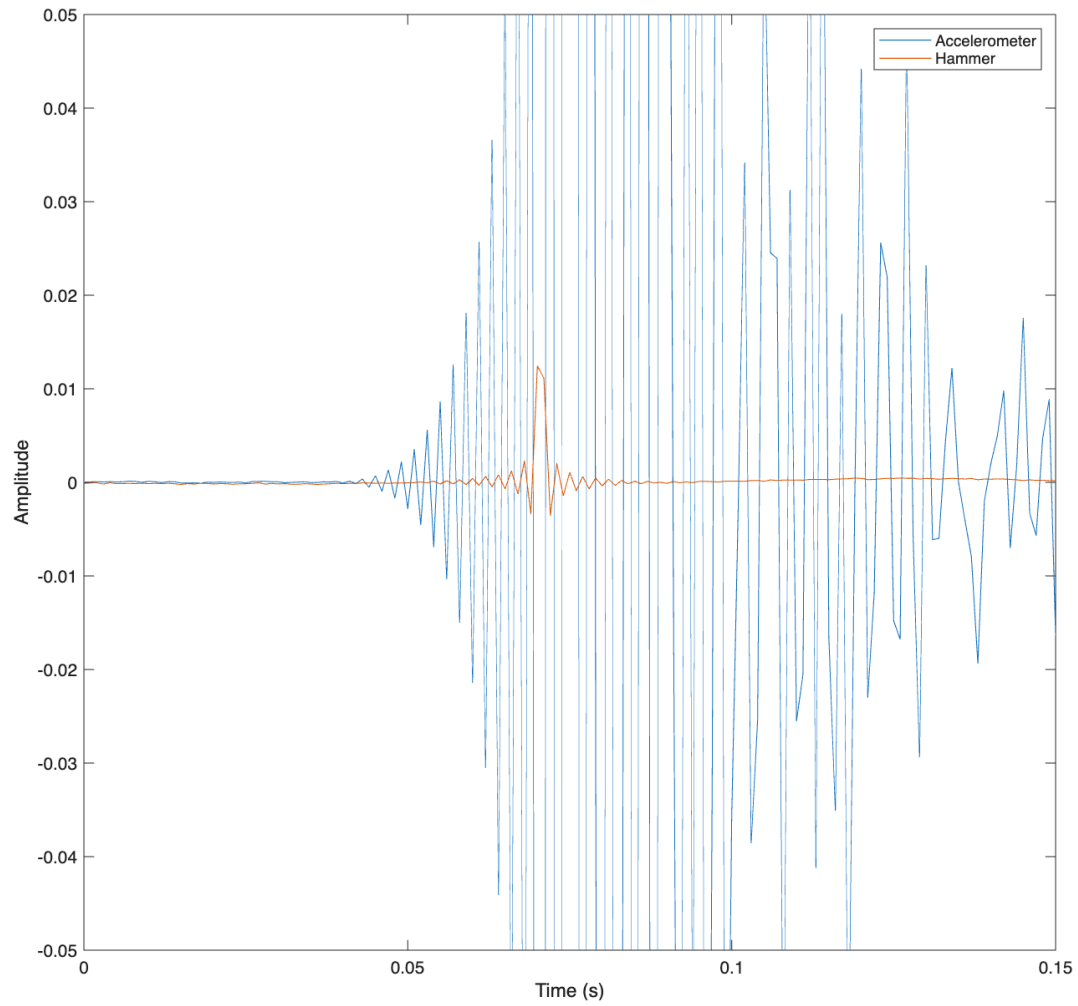


Figure 11: Raw Force and Acceleration Data

5.2 FFT

This is fast Fourier transform of the raw data. There is a lot of high frequency noise in the force.

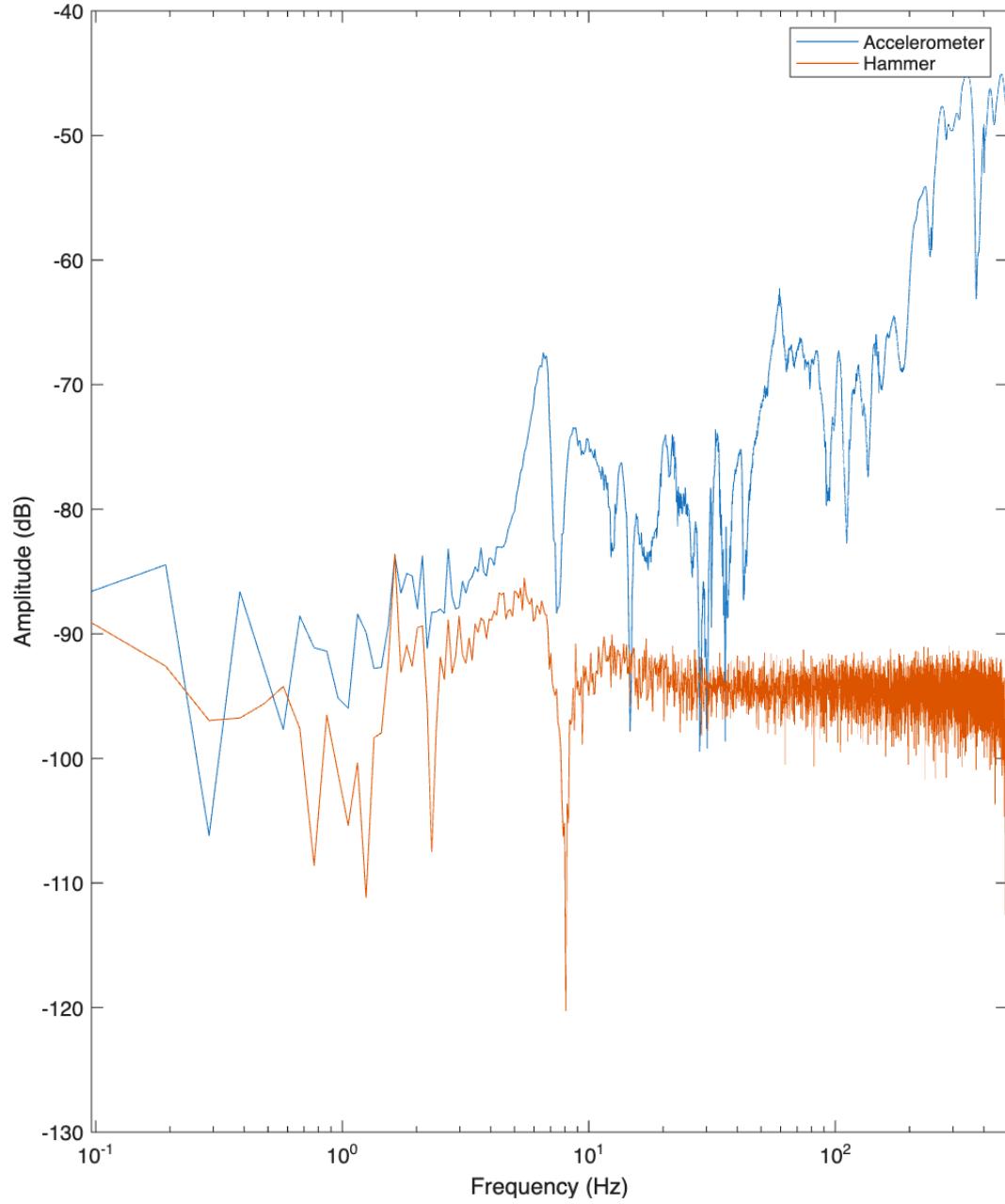


Figure 12: FFT of Force and Acceleration

5.3 Natural Frequencies

For all five nodes, the noise becomes amplified in the positive imaginary part of FRF making it impossible to identify the peaks to find the mode shapes.

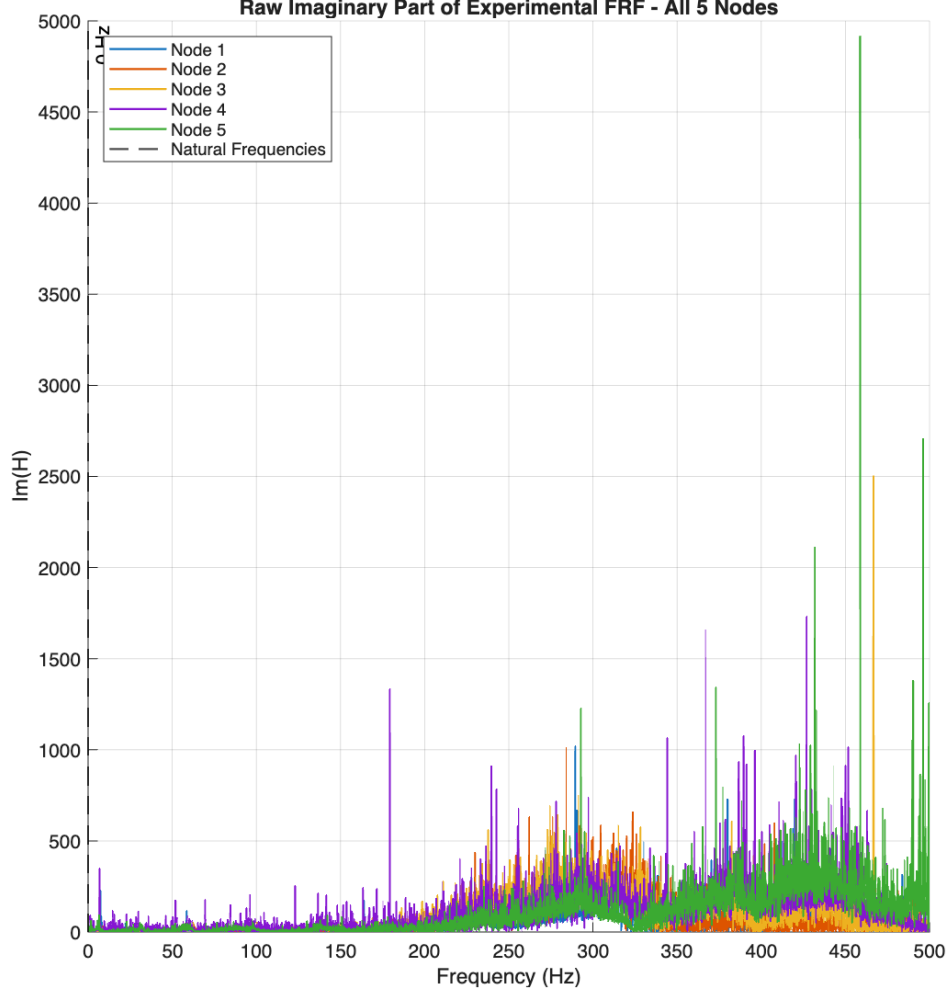


Figure 13: Positive Imaginary Part of the Forced Response Function

6 Discussion

6.1 Important Observations

6.1.1 Natural Frequencies

The natural frequencies show up as peaks on the imaginary part of the FRF because each mode component of it is most excited when $\omega = \omega_n$. This assumes that the inherent superposition from the other mode components influence the shape/positions of the peaks negligibly. See equation (14) for the exact details.

There are slight variations between each set of data (figure 8) of where the peaks occur because the beam does not behave exactly like the model, and there are slight inaccuracies stemming from human error and environmental noise. Human errors include not striking the beam at exactly an orthogonal angle to the beam or exactly in the center of the beam causing vibrations in the other axis of orientation. Furthermore, the beam is not actually

entirely fixed to the rest of the 3D printer, it still needed to slide vertically via screws, for nodes 1 and 5 are at the end of the beam therefore the boundary conditions could have affected it more, as well as the accuracy of the hammer strike.

6.1.2 Mode Shapes

The mode shapes illustrate how the system responds to the dominant modes/natural frequencies at the beam positions of interest. The mode shapes themselves are the superposition of sinusoids (with respective natural frequencies) with respect to the spatial mode components.

With regards to the mode shapes being presented in this report, figure 9 more or less speaks for itself. However, we could take note of a couple of things. For one, modes 1 and 2 both share (roughly) the same shape along the beam length, which makes sense given that their natural frequencies are close together in size (i.e. $261 \sim 274$). Mode 7, which is associated with the highest natural frequency, has the largest maximum amplitude, and is very biased toward one side of the beam. Mode 4 on the other hand has the lowest average amplitude of the bunch and it tends to oscillate closely to the beam's centerline. Overall, these modes characterize how the beam would respond to forces that resonate with the dominant natural frequencies.

6.2 Potential Reasons for Poor Secondary Measurements

The second set of impact tests was conducted with the accelerometer (offset by approximately one-quarter of the beam length from the center at node 3) in addition to the original centered position. However, the frequency response functions from these offset tests proved significantly noisier than the original dataset, with modal peaks that were difficult to identify and did not consistently align with those observed in the centered configuration, even after applying a pre-windowing and averaging out the FFT. The poor quality of the new data, possibly due to suboptimal accelerometer mounting, local structural effects, or minor degradation in sensor/hammer performance between the two trials, rendered it unsuitable for reliable mode shape extraction. Consequently, all natural frequency and mode shape results presented in this report were derived exclusively from the original high-quality centered-accelerometer dataset, while the offset results are retained here only to document the experimental challenges encountered.

6.3 Applications

Input shaping is now a common technology on most modern 3D printers, which come with accelerometers built-in to perform self-calibrations as necessary. In these applications the process typically involves a frequency sweep via the motion system (up to typically up to $100\text{--}200\text{Hz}$) and measuring the response. A sample power spectral density plot obtained from this process is shown in Figure 14.

However, a direct comparison of the data obtained in this experiment and a "input shaping" plot is difficult for a few reasons:

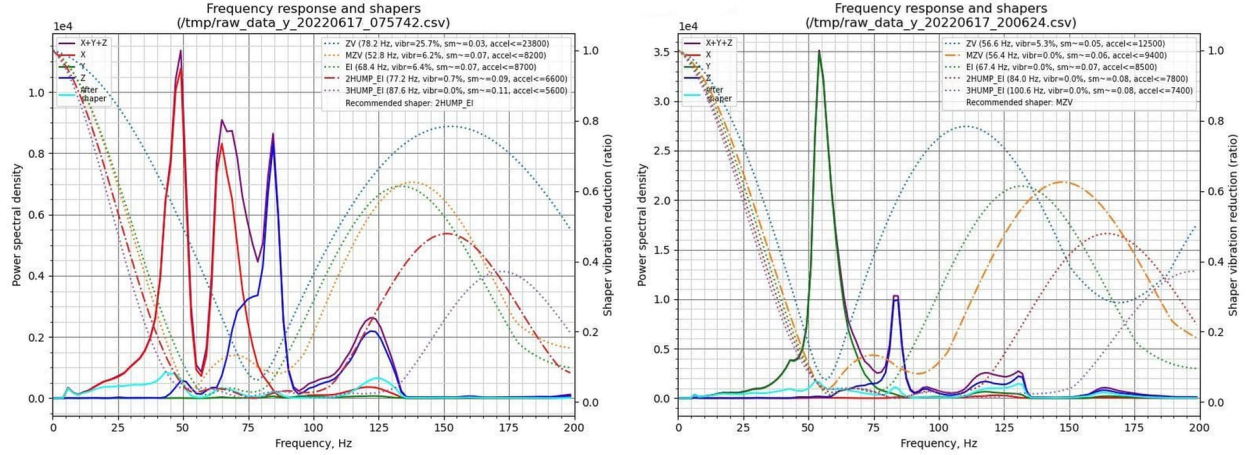


Figure 14: Sample PSD Plot

1. standard input shaping calibrations treat the printer as a lumped SDOF system²
2. the excitation is derived from the actual motion system (motors, timing belts, rails)
3. the frequency range is far narrower, only up to 200Hz (typ.)²

As such, a direct comparison of the data gathered in this experiment and widely available accelerometer data will be difficult. Future work could potentially perform an identical frequency sweep compared against publicly available values, or apply design changes based on the modal analysis to mechanically stabilize the beam for operating conditions, then test for print quality improvements.

7 Conclusion

In summary, the beam of the 3D printer was successfully analyzed using modal analysis. Although only 5 positions on the beam itself were measured against, the mode shapes provided a decent enough illustration as to how the beam vibrates in response to the dominant resonate frequencies. Additionally, although the dominant resonate frequency of mode 5 at the (most) rightward position measured failed to line up with the other modes at the same position, it is a single outlier. The other modes all (more or less) line up with each other in this manner, which give credence to the theoretical model, and modal analysis as a whole.

With all said, with the dominant resonate frequencies of the 3D printer analyzed and tabulated, future/present algorithms can be tuned to prevent the 3D printer from oscillating at these frequencies to avoid resonance in conjunction with print failures.

8 Appendix

8.1 MATLAB Code

```
%% Vibration Data Analysis - Peak Finding
clear; clc; close all;

% 1. Load the data for this specific point
data = load('Vibes - (.3) Beam - Tip 1.mat');
% load('Vibes - (.5) Beam center 3 - Tip 1.mat');
p = data.data;
amp_accel = p(:, 1); % Output (Beam Accelerometer)
amp_hammer = p(:, 2); % Input (Impact Hammer)
fs = data.fs;
N = length(p);
dt = 1 / fs;
t = (0:N-1).' * dt;
figure(1)
plot(t, data.data);
xlabel('Time (s)');
ylabel('Amplitude');
legend('Accelerometer', 'Hammer', 'FontSize', 18);
xlim([0, 0.15]);
ylim([-0.05, 0.05]);

%%

% -- RECALCULATE N, t, and f based on the new truncated length --
N = length(amp_hammer);
t = (0:N-1).' * dt;
df = fs / N;
f = (0:N-1).' * df;

figure(2)
% -- Create the Exponential Window (For the Accelerometer) --
% Smoothly force the "tail" of the ringing to zero to prevent leakage.
decay_rate = -log(0.01) / t(end);
exp_window = exp(-decay_rate * t);

% Apply the exponential window to the truncated accelerometer data
amp_accel_clean = amp_accel .* exp_window;
% Hammer is already truncated tightly around the peak, no window needed
amp_hammer_clean = amp_hammer;
```

```

% FFTs using the truncated and cleaned data
Y_accel = fft(amp_accel_clean) * dt;
Y_hammer = fft(amp_hammer_clean) * dt;

semilogx(f, 20 * log10(abs(Y_accel)), 'DisplayName', 'Accelerometer');
hold on;
semilogx(f, 20 * log10(abs(Y_hammer)), 'DisplayName', 'Hammer');
hold off;
xlim([0, fs/2]);
xlabel('Frequency (Hz)');
ylabel('Amplitude (dB)');
legend('FontSize', 18);

%%
H = Y_accel ./ Y_hammer;
figure(3)
semilogx(f, 20 * log10(abs(H)));
title('Experimental FRF (DB) vs FREQUENCY');
xlabel('Frequency (Hz)');
ylabel('Amplitude (dB)');
xlim([0, fs/2]);

f_half = f(1:floor(N/2));
H_half = H(1:floor(N/2));

figure(4);
plot(f_half, abs(imag(H_half)), 'b', 'LineWidth', 1.2);
xlabel('Frequency (Hz)');
ylabel('Amplitude');
title('Absolute Imaginary Part of Experimental FRF |Im(H)|');
grid on;

figure(5);
plot(f_half, (imag(H_half)), 'b', 'LineWidth', 1.2);

xlabel('Frequency (Hz)');
ylabel('Amplitude');
title('Imaginary Part of Experimental FRF |Im(H)|');
grid on;
%%
figure(6);
hold on;
legend_labels = cell(1, 5);
wns = [261, 274, 325, 363, 400, 430, 463];
%wns = [231,264,305,344,378,410,480]; % Natural frequencies array center

```

```

wns = [249,291,310,373,383,422,436,446];
for i = 1:5
    filename = sprintf('Vibes - (.%d) Beam - Tip 1.mat', i);

    data = load(filename);
    p = data.data;
    amp_accel = p(:, 1);
    amp_hammer = p(:, 2);
    fs = data.fs;
    N_raw = length(p); % Store original N for safety
    dt = 1 / fs;

    % Update variables for the truncated size
    N = length(amp_hammer);
    t = (0:N-1).' * dt;
    df = fs / N;
    f = (0:N-1).' * df;
    f_half = f(1:floor(N/2));

    % Calculate the FFT using the cleanly truncated data
    Y_accel = fft(amp_accel) * dt;
    Y_hammer = fft(amp_hammer) * dt;
    H = Y_accel ./ Y_hammer;
    H_half = H(1:floor(N/2));

    % Plot the results
    plot(f_half, abs(imag(H_half)), 'LineWidth', 1);
    legend_labels{i} = ['Node ', num2str(i)];

end

for k = 1:length(wns)
    label_str = sprintf('%d Hz', wns(k));
    if k == 1
        xline(wns(k), 'k--', label_str, 'LineWidth', 1, ...
            'LabelOrientation', 'aligned', 'LabelVerticalAlignment', 'top', 'FontSize', 14);
        legend_labels{end+1} = 'Natural Frequencies';
    else
        xline(wns(k), 'k--', label_str, 'LineWidth', 1, 'HandleVisibility', 'off', ...
            'LabelOrientation', 'aligned', 'LabelVerticalAlignment', 'top', 'FontSize', 14);
    end
end

% Formatting the overlaid plot

```

```

title('Raw Imaginary Part of Experimental FRF - All 5 Nodes');
xlabel('Frequency (Hz)');
ylabel('Im(H)');
xlim([200, fs/2]);
grid on;
legend(legend_labels, 'Location', 'northwest','FontSize', 18);

hold off;

%%
figure(15);

set(gcf, 'Position', [100, 100, 800, 1000]);
wns = [261, 274, 325, 363, 400, 430, 463];

for i = 1:5
    filename = sprintf('Vibes - (.%d) Beam - Tip 1.mat', i);
    data = load(filename);
    p = data.data;
    amp_accel = p(:, 1);
    amp_hammer = p(:, 2);
    fs = data.fs;
    dt = 1 / fs;
    N = length(amp_hammer);
    t = (0:N-1).' * dt;
    df = fs / N;
    f = (0:N-1).' * df;
    f_half = f(1:floor(N/2));

    Y_accel = fft(amp_accel) * dt;
    Y_hammer = fft(amp_hammer) * dt;
    H = Y_accel ./ Y_hammer;
    H_half = H(1:floor(N/2));

    subplot(5, 1, i);
    plot(f_half, imag(H_half), 'b', 'LineWidth', 1, 'DisplayName', 'Both Parts');
    hold on;
    plot(f_half, abs(imag(H_half)), 'r--', 'LineWidth', 1, 'DisplayName', 'Positive Only');

    % Add natural frequency lines to this specific subplot
    for k = 1:length(wns)
        if wns(k) > 0 % Prevents plotting a 0 Hz line when wns = 0
            % Removed the text label and added HandleVisibility off
            xline(wns(k), 'k--', 'LineWidth', 1, 'HandleVisibility', 'off');
        end
    end
end

```

```

end

title(['Node ', num2str(i)]);
ylabel('Im(H)');
xlim([200, fs/2]);
grid on;

if i == 5
    xlabel('Frequency (Hz)');
elseif i == 1
    legend('Location', 'Northwest');
end

hold off;
end

% Add a main title centered above all subplots
sgtitle('Experimental FRF - All 5 Nodes', 'FontWeight', 'bold', 'FontSize', 14);

%%

amplitudes = [
    0, -18.2729, -30.7332, 48.8158, -22.7461, -17.2999, 0;
    0, -13.4903, -25.3697, 38.1507, -20.6884, -14.8708, 0;
    0, -10.5152, -19.1198, 28.2759, 4.18415, 9.84516, 0;
    0, -5.2080, 7.5930, 13.4641, -1.31118, -7.65530, 0;
    0, 20.5031, 2.3801, 21.0836, -11.7431, -9.86541, 0;
    0, 27.7703, -3.5211, 23.3127, -25.4127, -11.8474, 0;
    0, 52.3374, 12.5647, 17.2164, -6.28191, 27.1918, 0
];
wns = [261, 274, 325, 363, 400, 430, 463];
nodes = 0:6;

% Plotting each mode in its own window
for i = 1:size(amplitudes, 1)
    figure(6 + i);
    plot([nodes(1), nodes(end)], [0, 0], 'k--', 'LineWidth', 1);
    hold on;
    plot(nodes, amplitudes(i, :), 'b-o', 'LineWidth', 1.5, 'MarkerFaceColor', 'b');
    hold off;
    title(sprintf('Mode %d (%d Hz)', i, wns(i)));
    ylabel('Amplitude');
    xlabel('Node Position');
    max_amp = max(abs(amplitudes(i, :)));
    ylim([-max_amp*1.2, max_amp*1.2]);
end

```

```

        xlim([-0.2, 6.2]);
end

%%
amplitudes = [
    0, -18.2729, -30.7332, 48.8158, -22.7461, -17.2999, 0;
    0, -13.4903, -25.3697, 38.1507, -20.6884, -14.8708, 0;
    0, -10.5152, -19.1198, 28.2759, 4.18415, 9.84516, 0;
    0, -5.2080, 7.5930, 13.4641, -1.31118, -7.65530, 0;
    0, 20.5031, 2.3801, 21.0836, -11.7431, -9.86541, 0;
    0, 27.7703, -3.5211, 23.3127, -25.4127, -11.8474, 0;
    0, 52.3374, 12.5647, 17.2164, -6.28191, 27.1918, 0
];
nodes = 0:6;

figure(16);
plot([nodes(1), nodes(end)], [0, 0], 'k--', 'LineWidth', 1.5, 'HandleVisibility', 'off')
hold on;

for i = 1:size(amplitudes, 1)
    plot(nodes, amplitudes(i, :), '-o', 'LineWidth', 1.5, 'MarkerSize', 6, 'DisplayName')
end
hold off;

title('All Experimental Mode Shapes');
xlabel('Node Position');
ylabel('Amplitude');
xlim([-0.2, 6.2]);
grid on;

% Increased the font size of the legend to 14
legend('Location', 'bestoutside', 'FontSize', 18);

```

References

- [1] Flashforge. *Adventurer 5M Pro 3D Printer*. URL: <https://www.flashforge.com/products/adventurer-5m-pro-3d-printer> (visited on 05/13/2026).
- [2] Sineos. *Interpreting the Input Shaper Graphs*. URL: <https://klipper.discourse.group/t/interpreting-the-input-shaper-graphs/9879> (visited on 05/13/2026).